



1.7

Rational Functions and End Behavior

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1

| ECHS Mathematics

| Teacher: Mohammad Abu-

Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: If numerator degree is less than denominator degree, the horizontal asymptote is $y = 0$.

2. Write one precise sentence explaining this principle: Equal degrees give a horizontal asymptote equal to the ratio of leading coefficients.

3. Write one precise sentence explaining this principle: If numerator degree is exactly one greater, polynomial division gives a slant asymptote.

4. Write one precise sentence explaining this principle: The remainder term divided by the denominator approaches 0 when the denominator degree exceeds the remainder degree.

Core

5. Find the horizontal asymptote of $\frac{5x^3-2}{2x^3+x}$.
6. Find the slant asymptote of $\frac{x^2+3x+5}{x+1}$.
7. Why may a rational graph cross a horizontal asymptote?
8. State what the formula $R(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{A(x)}{Q(x)}$ tells you and name any required restriction.
9. State what the formula $\deg A < \deg Q \Rightarrow R(x) - S(x) \rightarrow 0$ tells you and name any required restriction.
10. Create a second representation (table, graph, formula, or context) for one central relationship in Rational Functions and End Behavior. State its domain and justify one key feature.

Medium

11. Determine the end-behavior asymptote of $\frac{3x^2+1}{x^3-2}$.
12. Determine the end-behavior asymptote of $\frac{5x^3-x}{2x^3+7}$.
13. Determine the end-behavior asymptote of $\frac{x^2+4x+1}{x-1}$.

14. Determine the end-behavior asymptote of $\frac{2x^4+1}{x^2+3}$.

15. Determine the end-behavior asymptote of $\frac{-4x^5+x}{2x^5-1}$.

Hard

16. Divide $\frac{2x^3-3x^2+4}{x^2+1}$ and state the asymptote.

17. Find k so $\frac{kx^4+1}{2x^4-3}$ has horizontal asymptote $y = 5$.

18. Can a proper rational function have a nonzero horizontal asymptote? Explain.

19. Solve where $\frac{x^2+1}{x^2-1}$ crosses its horizontal asymptote.

Difficult

20. Find all k for which $\frac{x^2+kx+1}{x+1}$ has asymptote $y = x + 2$.

21. A rational function has polynomial asymptote $y = x^2 - 3x + 2$ and vertical asymptote $x = 1$. Construct one example.

22. Prove that if $R(x) - S(x) = A(x)/Q(x)$ with $\deg A < \deg Q$, then S is an end-behavior asymptote.

Challenge / HOT

23. Can a rational function have two different polynomial asymptotes as $x \rightarrow \infty$? Prove your answer.

24. Construct a rational function with slant asymptote $y = -2x + 1$ that crosses the asymptote at $x = 3$.

AP-Style Multiple Choice

25. $\frac{2x^3+1}{5x^3-4}$ has horizontal asymptote
(A) 0
(B) $2/5$
(C) $5/2$
(D) none
26. If numerator degree is one more than denominator degree, expect
(A) vertical asymptote only
(B) slant asymptote
(C) $y=0$
(D) a hole
27. Polynomial division writes a rational function as
(A) product only
(B) quotient plus remainder/denominator
(C) factor plus zero
(D) logarithm
28. An asymptote is
(A) a barrier never crossed
(B) a line/curve the graph approaches in end behavior
(C) always horizontal
(D) always in the domain
29. $\frac{x+1}{x^3+2}$ has end behavior
(A) $y=1$
(B) $y=x$
(C) $y=0$
(D) none
30. Equal degrees use
(A) ratio of constants
(B) ratio of leading coefficients
(C) difference of degrees
(D) sum of coefficients

AP-Style Free Response

31. Let $R(x) = \frac{2x^3+ax^2+bx+1}{x^2-1}$. Its slant asymptote is $y = 2x + 5$. (a) Find a and b. (b) Write division form. (c) State end behavior.

32. Construct a rational function with vertical asymptotes $x = -2, 3$, horizontal asymptote $y = 4$, and x-intercept 1. Explain each feature.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.